

Lab 4

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1. Automotive Tire Sales Modeling (2 points each part)

Using the data from TireSales.csv, answer the following questions.

```
Tires_df = read.csv("https://www.businessregression.com/Data/TireSales.csv")
```

- a. Fit a linear model to predict the response variable (Sales) from the variables: CompPrice, Income, Ads, Cars, Price and Age. Display a summary of the model.

```
reg = lm(Sales ~ CompPrice + Income + Ads + Cars + Price + Age, data = Tires_df)
summary(reg)
```

```
##
## Call:
## lm(formula = Sales ~ CompPrice + Income + Ads + Cars + Price +
##     Age, data = Tires_df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.0340 -1.2340  0.0302  1.3632  5.3442
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.6298913   1.2007986   2.190   0.0290 *
## CompPrice    0.0992546   0.0188199   5.274 2.00e-07 ***
## Income     -0.0003679   0.0029141  -0.126   0.8996
## Ads          0.0376981   0.0149658   2.519   0.0121 *
## Cars         8.0822521   0.1921791  42.056 < 2e-16 ***
## Price       -0.0966082   0.0153164  -6.308 6.31e-10 ***
## Age         -0.0157475   0.0129140  -1.219   0.2233
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.786 on 493 degrees of freedom
## Multiple R-squared:  0.7949, Adjusted R-squared:  0.7924
## F-statistic: 318.4 on 6 and 493 DF, p-value: < 2.2e-16
```

- b. Clearly state the regression equation.

$$\hat{y} = 2.62989 + 0.09925x_1 - 0.000368x_2 + 0.037698x_3 + 8.08225x_4 - 0.096608x_5 - 0.01575x_6$$

- c. Calculate the VIF values corresponding to each predictor variable.

```
library(car)
```

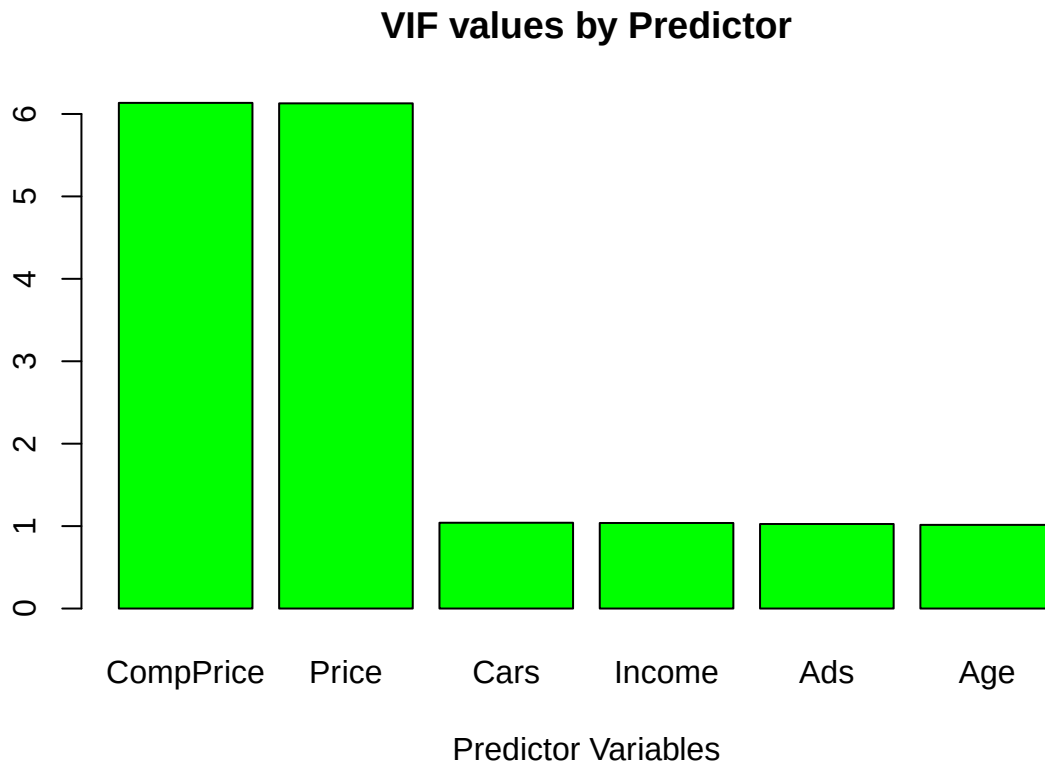
```
## Loading required package: carData
```

```
vif_values = vif(reg)
vif_values
```

```
## CompPrice    Income      Ads      Cars      Price      Age
##  6.134710  1.037864  1.025674  1.040806  6.128783  1.014830
```

d. Construct a bar graph using the VIF values calculated above.

```
sorted_vif_values <- vif_values[order(-vif_values)]
barplot(sorted_vif_values, col = "green", xlab = "Predictor Variables", main = "VIF values by Predictor")
```



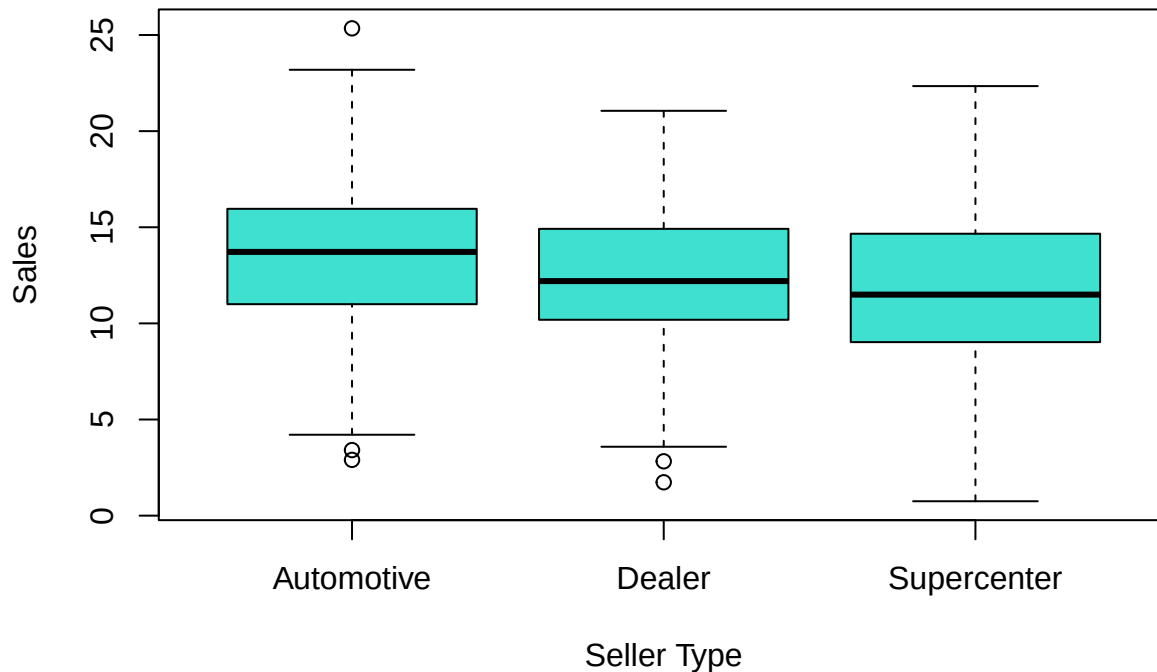
2. Automotive Tire Sales by Seller Type (2 points each part)

Using the data from `TireSales.csv`, answer the following questions.

a. Construct boxplots of the number of sales by seller type.

```
boxplot(Sales ~ SellerType, data = Tires_df, col = "turquoise", xlab = "Seller Type", ylab = "Sales", m
```

Boxplots of Number of Sales by Seller Type



- b. It is believed that sales are significantly different across the different seller types. Clearly state the null and alternative hypotheses.

$$H_O : \mu_1 = \mu_2 = \mu_3$$

$$H_A : \mu_1 \neq \mu_2 \neq \mu_3$$

- c. Calculate the F statistic.

```
reg2 = lm(Sales ~ SellerType, data = Tires_df)
anova(reg2)
```

```
## Analysis of Variance Table
##
## Response: Sales
##           Df Sum Sq Mean Sq F value    Pr(>F)
## SellerType  2  296.9  148.455   10.008 5.483e-05 ***
## Residuals 497 7372.7   14.834
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- d. Can you refute the null hypothesis using an F test with a significance level of $\alpha = 0.05$?

At significance level of $\alpha = 0.05$, we reject the null since the p -value < 0.05 .

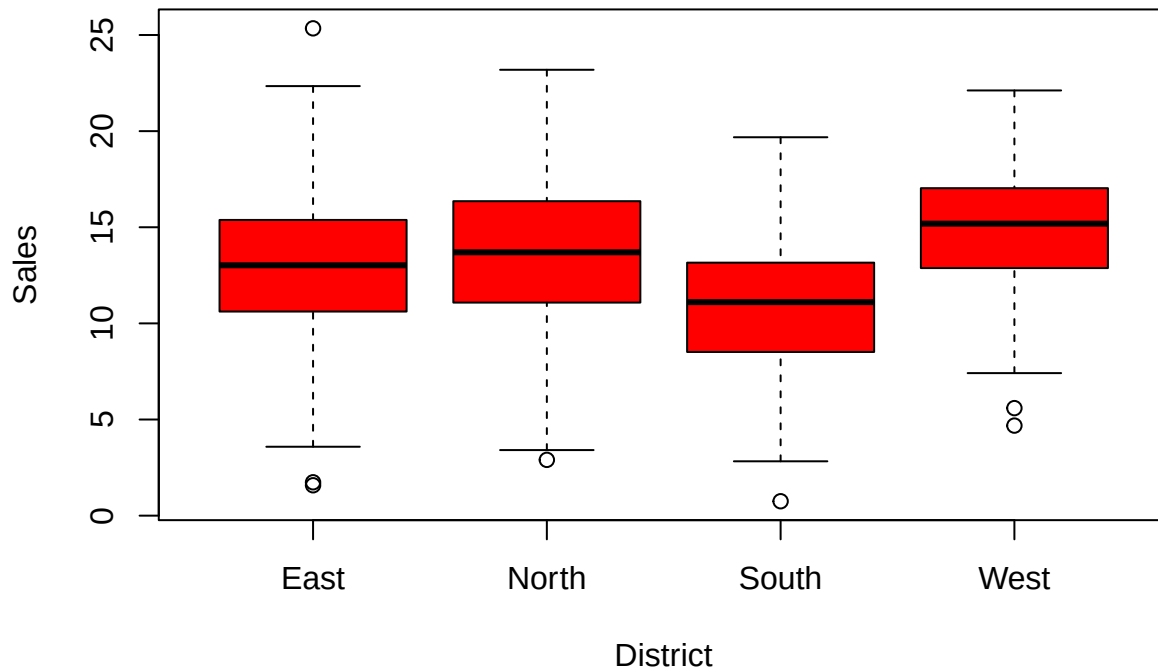
3. Automotive Tire Sales by District (2 points each part)

Using the data from `TireSales.csv`, answer the following questions.

- a. Construct boxplots of the number of sales by district.

```
boxplot(Sales ~ District, data = Tires_df, col = "red", xlab = "District", ylab = "Sales", main = "Number of Sales by District")
```

Number of Sales by District



- b. It is believed that sales are significantly different across the different districts. Clearly state the null and alternative hypotheses.

$$H_O : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_A : \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$$

- c. Calculate the F statistic.

```
reg3 = lm(Sales ~ District, data = Tires_df)
anova(reg3)
```

```
## Analysis of Variance Table
##
## Response: Sales
##      Df Sum Sq Mean Sq F value    Pr(>F)
## District    3   567.8   189.258   13.218 2.58e-08 ***
## Residuals 496  7101.8    14.318
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

- d. Can you refute the null hypothesis from the F test with a significance level of $\alpha = 0.05$?

At significance level of $\alpha = 0.05$, we reject the null since the p -value < 0.05

4. Coral Gables Housing (3 points each part)

Jigar is interested in purchasing a house in Coral Gables and wants to get the largest total living area for the price. He collects the size and sale price of 5 houses in a particular Coral Gables neighborhood. The data shown here reflect the size (in square feet) and the sale price (in thousands of dollars).

```
Size = c(2700, 3500, 2600, 5300, 4200)
Price = c(650, 875, 670, 1200, 981)
```

Use the Coral Gables housing data from the previous problems and do the following using R.

- a. Print a summary of the regression equation that predicts price as a function of house size.

```
reg4 = lm(Price ~ Size)
summary(reg4)
```

```
##
## Call:
## lm(formula = Price ~ Size)
##
## Residuals:
##      1      2      3      4      5
## -30.703  32.216   9.557  -7.466  -3.605
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 133.68052   45.10758   2.964  0.05937 .
## Size         0.20260    0.01188  17.048  0.00044 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 26.71 on 3 degrees of freedom
## Multiple R-squared:  0.9898, Adjusted R-squared:  0.9864
## F-statistic: 290.6 on 1 and 3 DF,  p-value: 0.0004397
```

- b. Use the regression equation to predict the price when the size of the house is 2500 square feet. **Hint:** `obs_x` has been given.

```
obs_x = data.frame(Size = 2500)
predict(reg4, data.frame(Size = 2500))
```

```
##      1
## 640.1829
```

- c. Find a 95% confidence interval for the price when the size of the house is 2500 square feet.

```
predict(reg4, data.frame(Size = 2500), interval = "confidence")
```

```
##      fit      lwr      upr
## 1 640.1829 582.1296 698.2362
```

- d. Find a 95% prediction interval for the price when the size of the house is 2500 square feet.

```
predict(reg4, data.frame(Size = 2500), interval = "prediction")
```

```
##      fit      lwr      upr
## 1 640.1829 537.2417 743.1241
```

5. Coral Gables Housing ANOVA (2 points each part)

Using the Coral Gables Housing data, do the following calculations without generating the ANOVA table (you will generate the ANOVA table in the next question).

- a. Calculate the *SSR*.

```
yhat = reg4$fitted.values
SSR = sum((yhat - mean(Price))^2)
SSR
```

```
## [1] 207370.2
```

b. Calculate the SST .

```
SST = sum((Price - mean(Price))^2)
SST
```

```
## [1] 209510.8
```

c. Calculate the MSR .

```
MSR = SSR/1
MSR
```

```
## [1] 207370.2
```

d. Calculate the F statistic.

```
n = length(Size)
SSE = SST - SSR
MSE = SSE/(n-2)
F = MSR/MSE
F
```

```
## [1] 290.6212
```

6. Coral Gables Housing ANOVA in R (3 points each part)

Using the Coral Gables Housing data, do the following using R.

a. Find the p-value of the F statistic calculated in the previous problem using the `pf` function.

```
1-pf(F,1,n-2)
```

```
## [1] 0.0004396693
```

b. Use the `anova` function to construct the ANOVA table.

```
anova(reg4)
```

```
## Analysis of Variance Table
##
## Response: Price
##           Df Sum Sq Mean Sq F value    Pr(>F)
## Size       1 207370  207370   290.62 0.0004397 ***
## Residuals  3   2141     714
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```